

Egocentric Hand Detection in Infant Video and One-Shot Object–Interaction Analysis

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CSCI B657 Computer Vision — Spring 2026

Code & data: [github.iu.edu/veravi/Infant_hands](https://github.com/iu-veravi/Infant_hands)

Abstract

Developmental psychology has traditionally observed infants from external cameras, missing the infant’s first-person view of objects and hands during play. Egocentric (head-mounted) cameras now make that view recordable, but standard computer vision pipelines fail on this footage: hands appear at unpredictable scales, perspective shifts constantly, and scenes are cluttered and uncontrolled. We present an end-to-end pipeline for **egocentric infant hand detection** and downstream **object–interaction analysis** across 23 toy classes. The pipeline (i) auto-labels frames with MediaPipe, (ii) corrects labels in CVAT, (iii) fine-tunes a YOLOv8m hand detector, and (iv) classifies the held object via DINOv2 prototype matching on a dynamically padded crop around each detected hand. On held-out Day-4 footage from five infants, the detector reaches Precision = 0.93, Recall = 0.93, mAP@50 = 0.96, mAP@50–95 = 0.72 (best mAP@50 = 0.967 at epoch 44). Per-infant interaction distributions are highly statistically distinct (Pearson $\chi^2 = 14,877.81$, dof = 88, $p < 10^{-3}$), recovering scientifically interpretable behavioral variation. We also report two **negative results**: (a) attempting to separate infant from caregiver hands using bounding-box geometry, and (b) using SAM2 as a standalone object proposer. Both failed for structural reasons that we diagnose in detail; the diagnoses motivated the final architecture.

1 Introduction

A central question in developmental psychology is *what infants see and touch* during everyday play, because what they see and touch is what they learn to name [1]. Conventional studies use external cameras placed in the room, but this third-person view systematically misses what is actually salient to the infant: the objects within reach, the toys held in the hand, and the moments of caregiver intervention. Egocentric (head-mounted) cameras worn by infants record exactly this first-person stream, but they introduce a hard computer-vision problem.

Egocentric infant footage is, in computer-vision terms, **adversarial**:

- Hands enter and leave the frame at extreme angles and scales.
- Frames are blurred by infant head motion.

- Scenes are cluttered domestic environments with no controlled lighting.
- Two-handed grasps and parent–infant joint manipulation produce overlapping, occluding hand instances.
- Toys are small (often < 100 px wide) and frequently partially hidden by the gripping hand.

The downstream scientific goal is **object–interaction analysis**: which toys does each infant grasp, when, and for how long? Reliable hand localization is a prerequisite. Reliable in-hand object identification is the next prerequisite.

Contributions. This paper makes three contributions:

1. A working hand-detection pipeline for egocentric infant video, reaching $\text{mAP@50} = 0.96$ on held-out footage from five infants over four days, with stable convergence and no overfitting.
2. A one-shot object-interaction labeling head that uses DINOv2 prototype matching on a hand-anchored crop, requiring no per-class supervised training, and producing per-infant interaction statistics that are highly statistically distinct across subjects ($p < 10^{-3}$).
3. Two negative results with full diagnostics: hand-size-based identity discrimination and SAM2-as-proposer both failed for structural reasons we identify and explain. These failures shaped the final architecture.

Research questions.

RQ1 Can a YOLO-class detector trained on auto-labeled, human-corrected egocentric infant frames generalize across infants and across days?

RQ2 Can in-hand object identity be recovered without per-class supervised training, using a frozen general-purpose feature embedder?

RQ3 Can infant-vs-caregiver hand identity be recovered from bounding-box geometry alone?

The answers, in order, are: yes; yes (for 23 toy classes, with caveats on localization precision); and no, structurally—discussed in §5.

2 Background and Related Work

Egocentric video and developmental psychology. Bambach et al. [1] established that toddler-egocentric video, treated as a curriculum of close-up, head-stabilized object views, dramatically improves visual object learning compared with adult-egocentric or third-person video. That result motivates the present project: if egocentric infant footage *can* support object learning, automated tools to extract *what is in the infant’s hand* unlock new analyses across larger cohorts than manual coding allows.

Hand detection. Skin-color and depth-based hand detection give way to learned detectors in cluttered scenes. **MediaPipe Hands** [7] provides 21-keypoint landmarking but is brittle on egocentric video; we use it only as a label-bootstrapping signal. **YOLOv8** [2] is the current default real-time detector and the workhorse of our pipeline. We chose YOLOv8m as the medium-capacity tradeoff between accuracy and inference latency on a single A100.

Vision foundation models. DINOv2 [3] provides self-supervised ViT features that transfer to dense prediction and retrieval without task-specific fine-tuning. We use `vit_base_patch14_dinov2` (via `timm`) at 518×518 input as a frozen embedder. SAM2 [4] extends Segment Anything to video with hierarchical mask generation; we evaluate its `SAM2AutomaticMaskGenerator` as a class-agnostic proposer in §5.3.

One-shot / prototype-based classification. Per-class prototype embeddings + cosine matching is a long-standing few/one-shot recipe [6]. Its appeal here is the absence of object-level supervised data: we need only ~ 30 reference frames per toy class to construct the prototype bank.

The gap our work fills. Prior work has either (a) trained supervised object detectors on egocentric data with extensive per-class annotation, or (b) used prototype matching on clean, centered, third-person images. Our combination—*YOLO-detected hand box as a learned spatial prior + DINOv2 prototype matching on the padded crop*—addresses the egocentric-specific failure of class-agnostic proposers (§5.3) without requiring per-class object annotation.

3 Methods

Figure 1 shows the end-to-end pipeline: a one-time *train phase* (top row) that produces YOLOv8m hand-detector weights, and a re-runnable *inference phase* (bottom row) that uses those weights together with frozen DINOv2 prototype matching to label hand-region object interactions in held-out video.

End-to-end pipeline: egocentric video → hand detection → one-shot object-interaction labeling

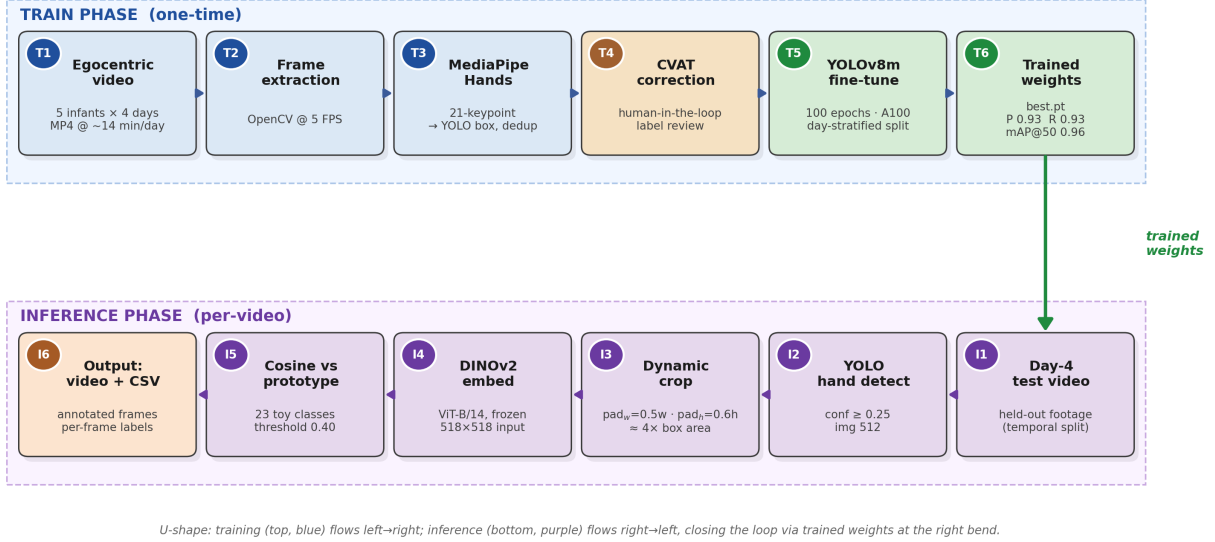


Figure 1: End-to-end pipeline. *Top row (train phase):* egocentric MP4 → frame extraction at 5 FPS → MediaPipe auto-labels → CVAT human-in-the-loop correction → YOLOv8m fine-tune → trained weights (P=0.93, R=0.93, mAP@50=0.96). *Bottom row (inference phase):* held-out Day-4 video → YOLO hand detection (conf ≥ 0.25) → dynamic crop (~4× box area) → DINOv2 ViT-B/14 embedding at 518×518 → cosine similarity vs. 23-class prototype bank (threshold = 0.40) → per-frame CSV and annotated video. The green arrow denotes weights produced once and consumed at inference.

3.1 Dataset

The corpus consists of **5 infant subjects** (IDs 25131, 25138, 25176, 25190, 25602) each recorded over **4 days** with a head-mounted egocentric camera. Behavioral annotations are stored in `Spatial_Master_5_subj` (onset/offset timestamps in milliseconds; `child_in_hand._` and `parent_in_hand._` contact labels). Source video paths follow `{DATA_ROOT}/{subject}/MP4/{subject}c_day{N}_merged.mp4`.

We held out **Day 4** of every subject as a test set. Days 1–3 were used for training and validation (Day 3 → val, Days 1+2 → train), using a **day-stratified split** rather than a random per-frame split to avoid temporal data leakage.

3.2 Hand-detection annotation pipeline (Stages 1–4)

Frames were extracted from each MP4 at **5 FPS** using OpenCV (`hands.py`). MediaPipe Hands (`max_hands=2`, `min_detection_confidence=0.4`) provided initial 21-keypoint landmarks, which were converted to YOLO-format bounding boxes (`generate_hand_labels.py`). Boxes were clamped to `[0, 1]` to prevent out-of-frame coordinates, zero-area boxes were dropped, and duplicates within a frame were removed using `order-preserving dict.fromkeys()`.

The auto-generated labels were then **manually reviewed and corrected in CVAT** [5]: missed hands added, false positives removed, padding tightened, and boxes adjusted around occlusion. Without this human-in-the-loop step, MediaPipe alone produced too many systematic errors near frame edges and at extreme angles.

3.3 YOLOv8m fine-tuning (Stage 6)

We fine-tuned `yolo8m.pt` on the corrected labels with the configuration in Table 1.

Table 1: YOLOv8m fine-tuning hyperparameters. Source: `scripts/config.py`.

Hyperparameter	Value
Epochs	100
Image size (train)	640
Image size (inference)	512
Batch size	128
Workers	8
Optimizer	YOLOv8 default (SGD, auto-tuned)
Train hardware	NVIDIA A100 (BigRed200)
Random seed	42

The day-stratified split, `SPLIT_SEED = 42`, and `sorted(os.listdir())` everywhere ensure reproducibility.

3.4 Object–interaction labeling (DINOv2 prototype matching)

Prototype construction. For each of 23 object classes (Figure 2; the `Bed` class is excluded at load time because it produced spurious matches), we curated ~ 30 reference frames showing the object in a representative pose. Each frame was resized to 518×518 , normalized with mean/std $(0.5, 0.5, 0.5)$, and embedded with DINOv2 (`vit_base_patch14_dinov2` from `timm`, frozen). Per-class embeddings were averaged and ℓ_2 -normalized, producing a single prototype vector per class. Prototypes are stored in `prototypes.pth`.

DINOv2 prototype reference images — 23 object classes

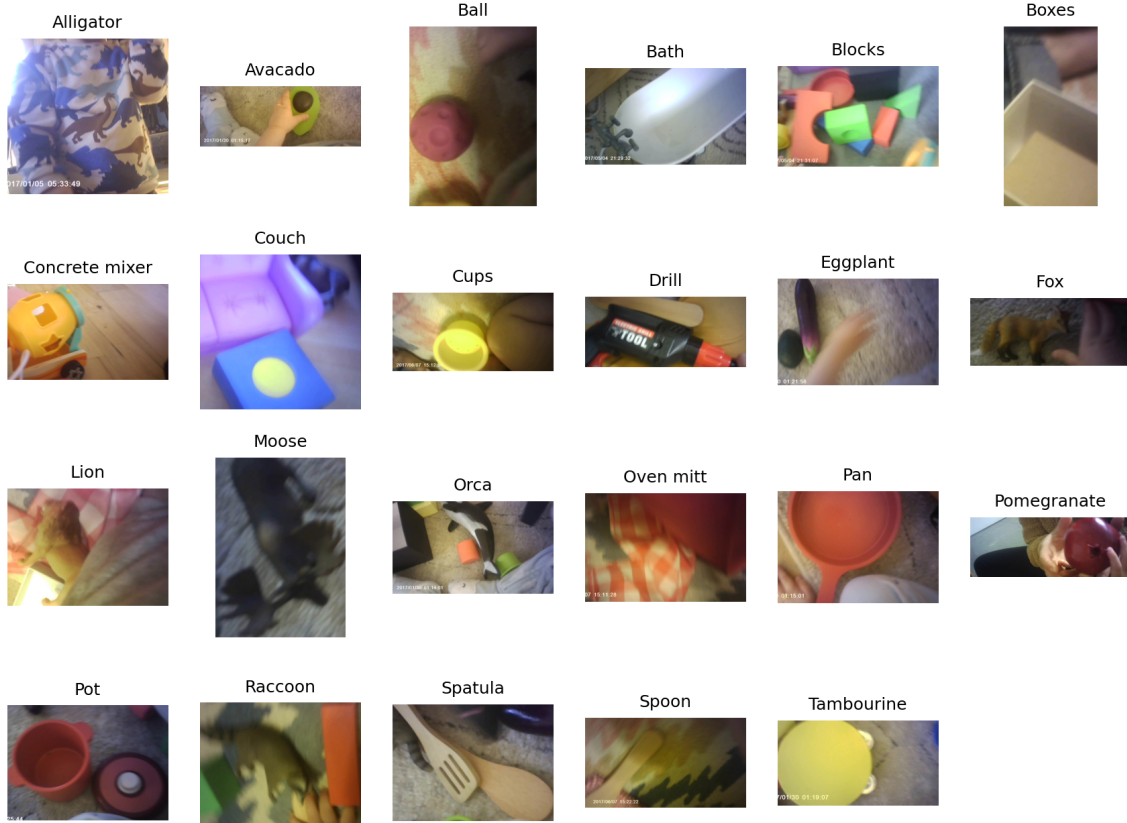


Figure 2: The 23 object classes in the DINOv2 prototype bank, one example image per class drawn from `toys_selected/`. The `Bed` class is excluded at inference time. Prototypes are constructed by averaging ~ 30 DINOv2 embeddings per class and ℓ_2 -normalizing.

Inference (production pipeline, `scripts/final.py`). For each video frame:

1. Run YOLOv8m $\rightarrow$ set of hand bounding boxes with confidences.
2. For each hand box (x_1, y_1, x_2, y_2) with width w , height h , expand into a hand-anchored object crop with dynamic padding:

```
pad_w = int(0.5 * w) # +50% on each side
pad_h = int(0.6 * h) # +60% on each side
```

This produces a crop $\sim 2\times$ the hand-box width and $\sim 2.2\times$ the height ($\approx 4\times$ area), reliably containing the held object. Crops below `CROP_MIN_SIZE` = 50 px are discarded.

3. Resize the crop to 518×518 , embed with DINOv2, ℓ_2 -normalize.
4. Compute cosine similarity against every prototype; take the arg max.
5. If the best score is below `DINO_THRESHOLD` = 0.4, label `unknown`.

6. Render annotated frames (green hand box, blue object box, class label) and write a per-frame CSV row: (frame, time_sec, hand_id, hand_conf, object_label, object_conf).

The pipeline is class-agnostic at the cropping stage and only becomes class-specific at the cosine-match stage. Adding a new toy class requires only ~ 30 new reference images and a one-time prototype rebuild—no retraining.

3.5 Compute and reproducibility

All training and large-batch inference ran on the BigRed200 HPC cluster (NVIDIA A100). SLURM job scripts are in `slurm/`; local equivalents in `bash/`. The single source of truth for all hyperparameters and paths is `scripts/config.py`, which auto-detects HPC vs. local environments.

4 Results

4.1 Hand-detection metrics (RQ1)

Training dynamics. Figure 3 shows the 100-epoch training curves for box loss, classification loss, precision/recall, and mAP. Both losses decrease monotonically, with the validation curves tracking training closely throughout, indicating no overfitting. Precision and recall stabilize above 0.90 by epoch ~ 30 and remain stable; mAP@50 plateaus at ≈ 0.96 from epoch ~ 40 onward, with the best epoch (44) achieving mAP@50 = 0.967 and mAP@50–95 = 0.685.

YOLOv8m hand-detection training (100 epochs)

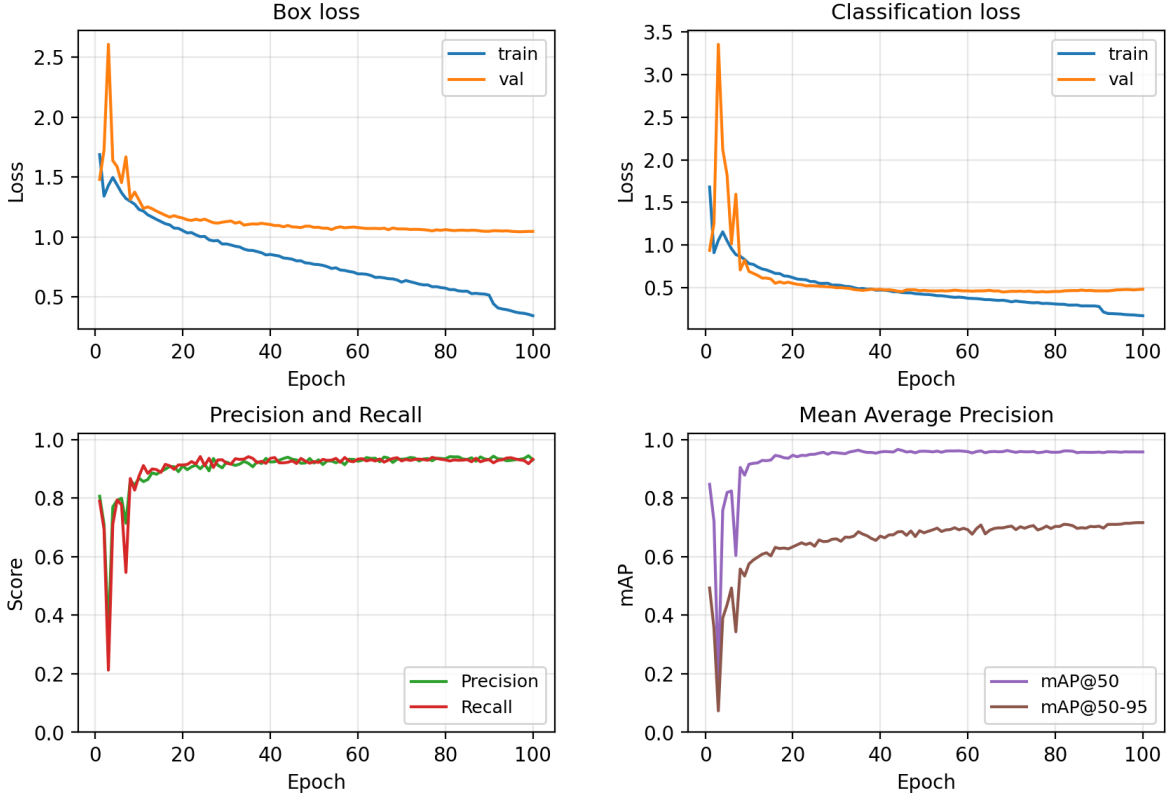


Figure 3: YOLOv8m fine-tuning over 100 epochs on auto-labeled, human-corrected egocentric infant frames. Top row: train and validation losses (box and classification). Bottom row: precision/recall and mean average precision at IoU thresholds 0.50 and 0.50–0.95. Validation losses track training losses, indicating no overfitting; metrics plateau by epoch ~ 40 .

Final test metrics. Table 2 reports YOLOv8m metrics at the final training epoch on the validation set drawn from Day 3 (held out from training). Confusion matrix and qualitative sample predictions are shown in Figures 4 and 5 respectively.

Table 2: Hand-detection performance on the validation set. *Best* is the best epoch by mAP@50 across the 100-epoch run.

Metric	Final epoch	Best epoch (44)
Precision	0.931	0.940
Recall	0.933	0.923
mAP@50	0.958	0.967
mAP@50–95	0.717	0.685

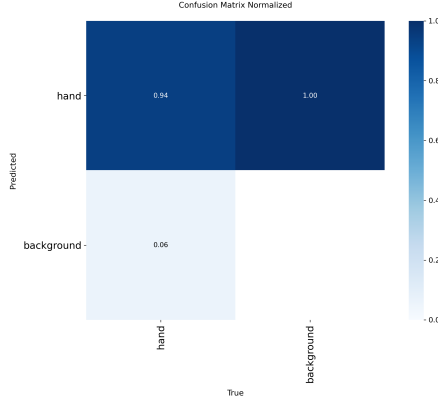


Figure 4: Normalized confusion matrix on the validation set. The single-class hand detector cleanly separates the *hand* class from background.

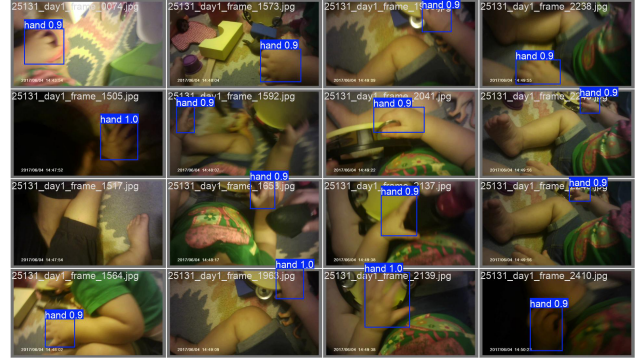


Figure 5: Sample validation predictions on Day-3 held-out frames. Bounding-box confidences are typically ≥ 0.9 , including under cluttered backgrounds and partial occlusion.

Day-4 inference confidence. Across all 5 infants on the Day-4 held-out test set, the pipeline emits 40,689 hand detections and 15,463 of them receive an object label above the DINO threshold. Figure 6 shows the distributions. The YOLO confidence distribution is heavily right-skewed: median hand-detection confidence is 0.86 and the modal bin is 0.93–0.96, indicating that the detector is decisive on most hands, with a long tail of marginal detections near the 0.25 threshold. The DINO cosine-similarity distribution is approximately exponential above the 0.40 threshold, with a median of 0.50 — consistent with one-shot prototype matching on out-of-distribution egocentric crops.

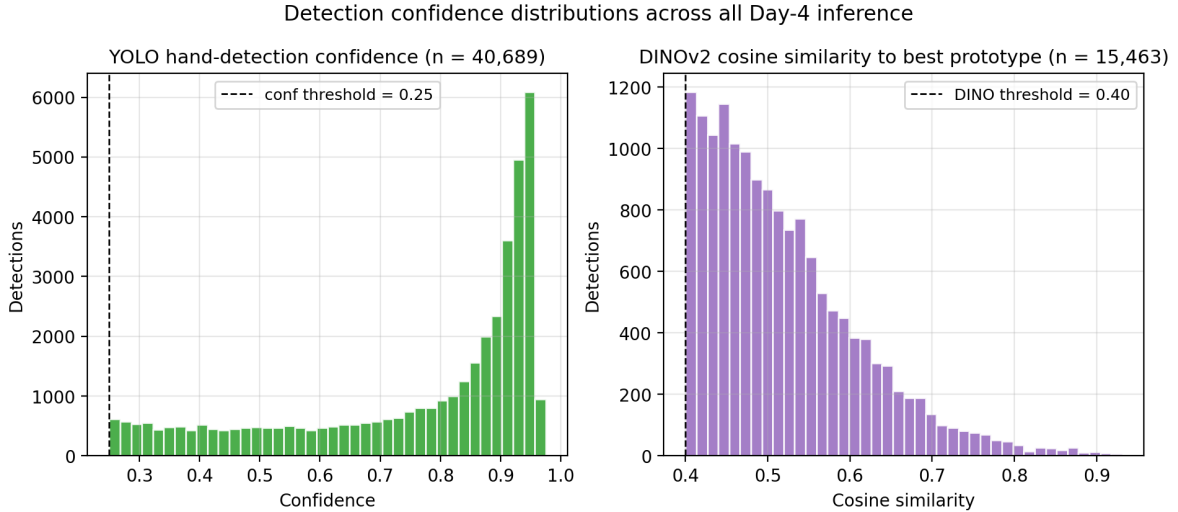


Figure 6: Confidence distributions across all Day-4 inference (5 infants, $\sim 40K$ detections). Left: YOLO hand-detection confidence is right-skewed with mode at ~ 0.95 . Right: DINOv2 best-prototype cosine similarity is approximately exponential above the 0.40 threshold (median 0.50, max ~ 0.9).

4.2 Object–interaction patterns (RQ2)

Across the 5 subjects on Day 4, the pipeline emits hand-region object labels for between 230 and 6,562 hand-frames per infant. Figure 7 shows the proportional breakdown of the 23 object classes per infant. The distributions are visibly distinct:

- **Baby 25131** ($n = 2,198$): cup-dominated ($\approx 63\%$ Cups), with substantial Boxes ($\approx 8\%$).
- **Baby 25138** ($n = 1,288$): spoon-dominated ($\approx 53\%$ Spoon), with notable Lion and Eggplant interaction.
- **Baby 25176** ($n = 5,185$): cup-dominated ($\approx 51\%$ Cups), with diverse secondary classes (Pot, Pan, Avocado, Lion).
- **Baby 25190** ($n = 230$): smallest sample; bath/spoon-mixed ($\approx 36\%$ Spoon, $\approx 28\%$ Cups). The small sample size limits the strength of single-class claims.
- **Baby 25602** ($n = 6,562$): the most diverse profile; Blocks-led ($\approx 27\%$), with substantial Avocado, Pomegranate, Pot, and Alligator components.

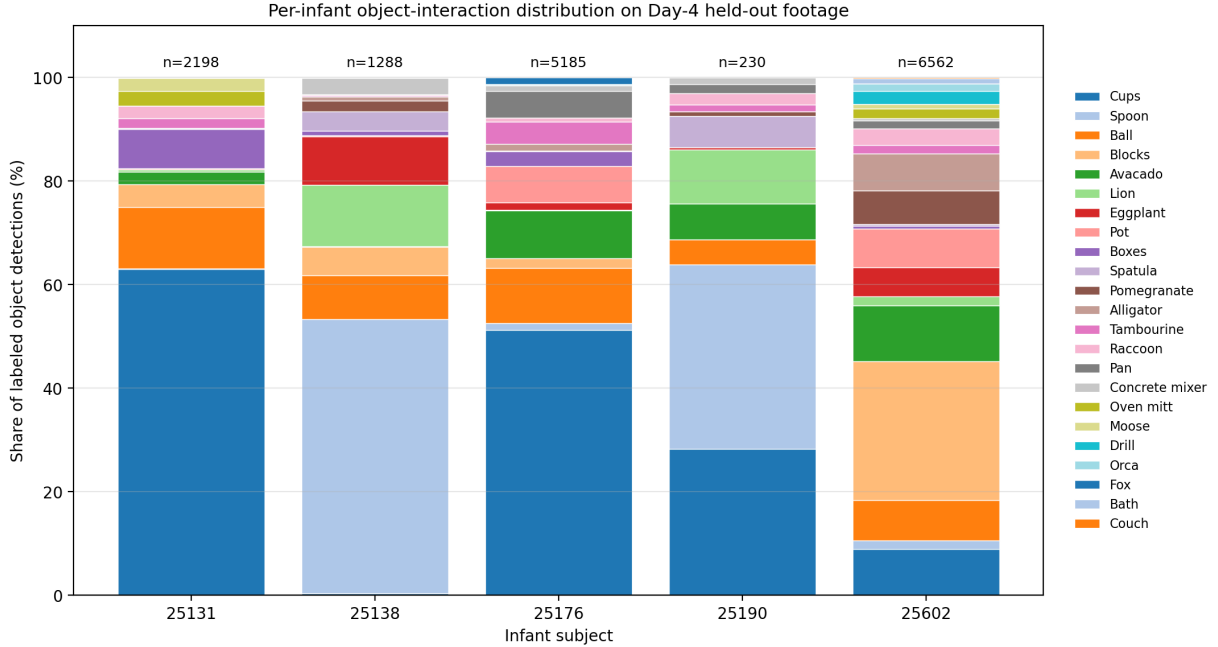


Figure 7: Per-infant Day-4 object-interaction distribution. Stacked bars show the proportional breakdown of DINOv2 one-shot object detections for each infant (legend: 23 toy classes). n above each bar is the number of detections that received a class label above the DINO threshold (**unknown** excluded). The distributions are clearly distinct across infants; the difference is statistically significant (Pearson $\chi^2 = 14,877.81$, $\text{dof} = 88$, $p < 10^{-3}$).

Statistical test and effect size. Treating each infant $\times$ object-class count as a contingency cell on the 23×5 contingency table (after pooling rare classes with row sum < 5 , 23 classes retained), the null hypothesis that interaction-class distribution is independent of infant identity is rejected by a Pearson χ^2 test of independence: $\chi^2 = 14,877.81$, $\text{dof} = 88$, $p < 10^{-3}$. To complement the test,

we report **Cramér’s V** $= \sqrt{\chi^2/[N \cdot (\min(r, k) - 1)]} = 0.490$ on the same table. Cramér’s V values > 0.25 are conventionally considered a large effect, so the per-infant variation is both statistically significant and practically large, not a knife-edge significance produced by a large N .

Per-infant diversity and unknown rate (Table 3). Per-infant Shannon entropy of the labeled-class distribution ranges from 2.00 bits (Baby 25131, cup-dominated) to 3.62 bits (Baby 25602, the most diverse profile) — a 1.6-bit spread, equivalent to a $\sim 3\times$ difference in effective number of classes used. The fraction of YOLO-detected hands that fail to match any prototype above threshold (**unknown**) varies from 46% (Baby 25602) to 93% (Baby 25190): the high-unknown infants tend to be those with smaller absolute detection counts, suggesting either fewer in-frame toys or a sample-size effect on prototype matching. We treat the **unknown** rate as a calibration signal — it is reported here rather than silently dropped (cf. Limitations §5).

Table 3: Per-infant diversity and **unknown** rate on Day-4 inference. *Entropy*: Shannon entropy of the labeled-class distribution (bits). *#classes*: number of distinct toy classes the pipeline assigned with score above threshold. *Top class*: most-detected class with its share of labeled detections.

Infant	n_{total}	n_{labeled}	unknown %	entropy (bits)	top class
25131	9,628	2,198	77.2%	2.00	Cups (63.0%)
25138	4,670	1,288	72.4%	2.37	Spoon (53.0%)
25176	10,971	5,185	52.7%	2.60	Cups (51.2%)
25190	3,320	230	93.1%	2.59	Spoon (35.7%)
25602	12,100	6,562	45.8%	3.62	Blocks (26.8%)

Per-infant qualitative detection examples. Figure 8 shows one representative annotated Day-4 frame per infant, drawn directly from the YOLO+DINO output videos in `data/yolo_outputs_new/final/`. Each panel renders the live pipeline output: a green YOLO hand bounding box with a confidence score, an enclosing blue box for the dynamically padded DINO crop, and the predicted object class above. Briefly, frame-by-frame:

- **Baby 25131** (frame 9,215, $t = 307$ s): two simultaneous hand detections, both crops classified as *Cups*. The pipeline correctly handles two-handed manipulation of a translucent yellow cup.
- **Baby 25138** (frame 8,747, $t = 292$ s): a single hand detection with crop labeled *Spoon*. The infant grips a plastic utensil; the YOLO box is tight on the hand and the padded crop reaches the held spoon.
- **Baby 25176** (frame 9,414, $t = 314$ s): two-handed *Cups* manipulation, again with both hands picking up the same prototype, consistent with this infant’s cup-dominated Day-4 profile (51% *Cups*).
- **Baby 25190** (frame 16,893, $t = 563$ s): two hands detected with the bath-time spoon visible; this infant’s footage has the smallest sample ($n = 230$), reflecting fewer object-interaction frames overall.
- **Baby 25602** (frame 12,890, $t = 430$ s): two-handed manipulation of a green plush *Alligator*. This infant has the most diverse interaction profile of the cohort.

Representative Day-4 detection frames (one per infant): green = YOLO hand box, blue = padded DINO object crop with class label

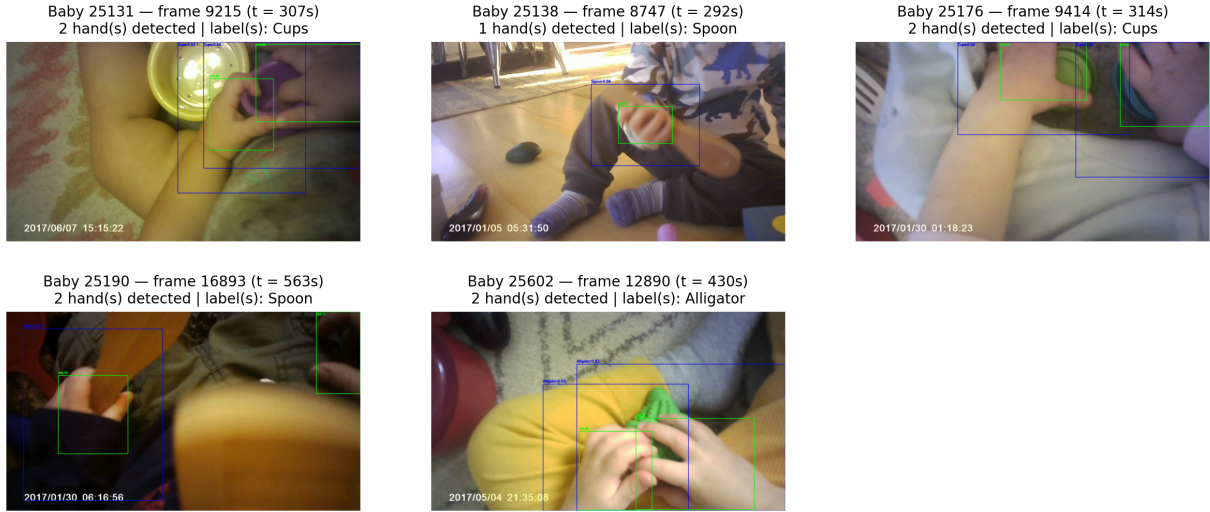


Figure 8: Representative Day-4 detection frames, one per infant. Green: YOLO hand bounding box with confidence. Blue: padded crop fed to DINOv2 (with predicted object class above each pair). Frames are drawn from the YOLO+DINO output videos in `data/yolo_outputs_new/final/`; selected as the median-by-time frame among the top 50 frames per infant ranked by simultaneous-hand count and YOLO confidence. The pipeline successfully labels both single- and two-handed manipulations across infants and toy classes.

Temporal structure. Figure 9 shows hands-per-frame timeseries for each infant across the full Day-4 session. Hand presence is **bursty**: long intervals of zero or one hand are punctuated by short two-hand episodes (object manipulation, give-take with caregiver). Reading the panels:

- Baby 25131 (~ 825 s session): a relatively flat low-activity baseline, with visible bursts around 100–150 s and 300–450 s.
- Baby 25138 (~ 530 s, the shortest session): activity ramps up after ~ 30 s and stays elevated, consistent with the infant being actively engaged with one toy class (Spoon).
- Baby 25176 (~ 857 s): the most variable temporal profile, with sustained two-handed activity from ~ 250 –500 s and another burst around 600 s.
- Baby 25190 (~ 760 s): mostly single-hand baseline with a clear two-handed episode around 250–280 s; consistent with the small ($n = 230$) labeled-detection count.
- Baby 25602 (~ 960 s, the longest session): dense early bursts from ~ 50 –250 s, sparser late-session activity.

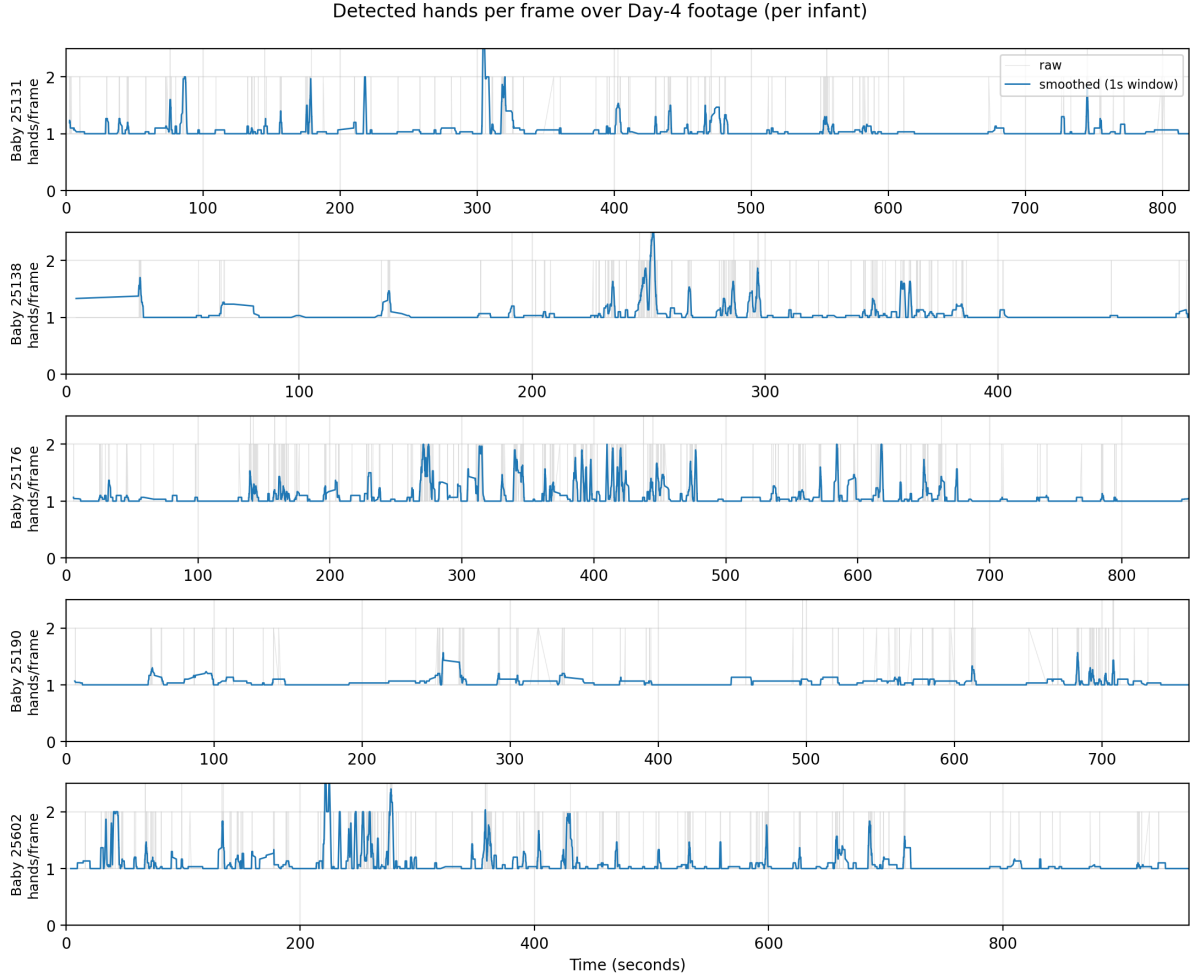


Figure 9: Hands-per-frame timeseries for each infant on Day 4. Light gray: raw per-frame counts. Blue: 1-second smoothed envelope. Hand presence is bursty across all infants, but the timing, density, and duration of two-handed episodes differs by infant. The bursts visible here align with the manipulation episodes captured in Figure 8.

Qualitative hand crops. Figure 10 shows a sample of YOLO-detected hand crops across infants and days. The detector recovers hands across a wide range of poses, scales, lighting, occlusion levels, and backgrounds.

YOLO-detected hand crops (sample across subjects and days)

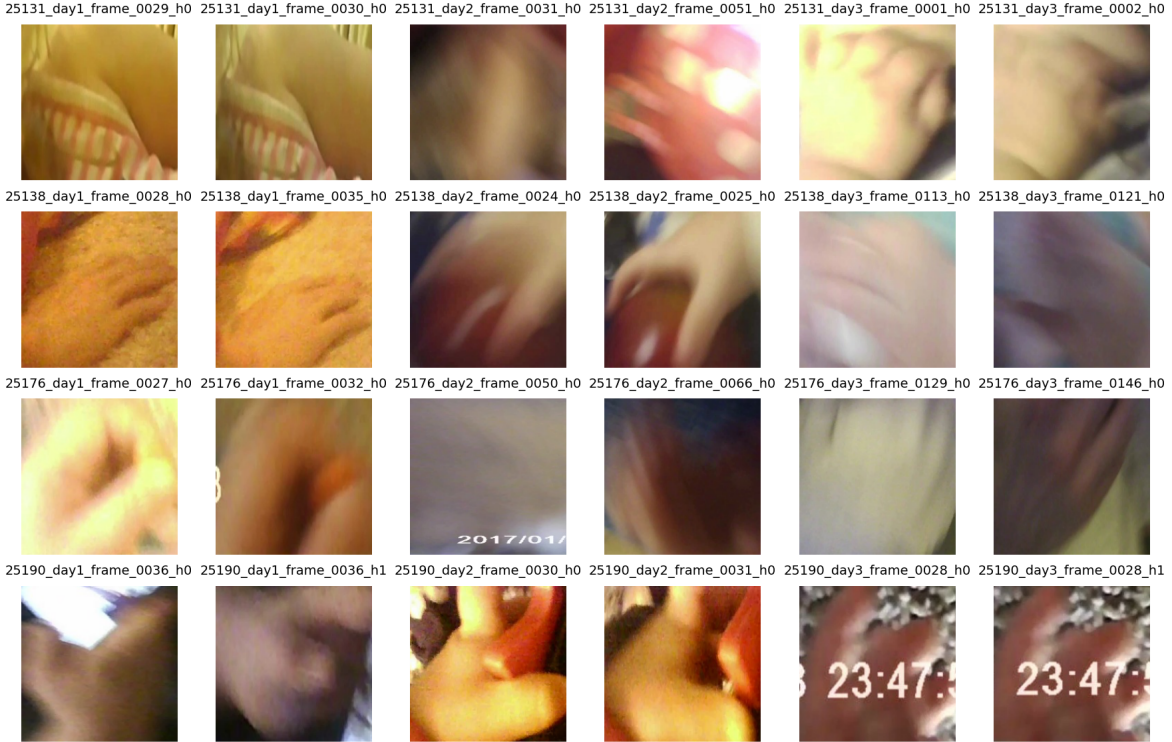


Figure 10: Sample of 24 YOLO-detected hand crops drawn across infants and days. The detector finds hands across pose, scale, occlusion, and lighting variation. These crops are the inputs (after dynamic padding) to the DINOv2 prototype-matching head.

4.3 Generalization

The detector was evaluated on Day-4 footage that is both *temporally separated* from training (Day 4 vs. Days 1–2) and, for a separate **data_freeplay** test set, *environment-shifted* (different room layouts and toys). YOLO confidence on the new-environment test set remains qualitatively comparable to in-distribution performance (Figure 5 shows in-distribution; the **data_freeplay** videos are produced by the same model without retuning). DINO prototype matching did not require any environment-specific tuning because the prototypes are environment-agnostic embeddings.

5 Discussion

5.1 What worked, and why

The combination of (i) a learned domain-specific spatial prior—the YOLO hand detector—and (ii) a frozen general-purpose feature embedder—DINOv2—is what allowed us to do object-interaction labeling without per-class supervised training. The hand detector tells us *where* the interaction is; DINOv2 tells us *what* is in the crop; the prototype bank tells us *which class* of object that “what” corresponds to. Each component is replaceable: if a stronger embedder or detector arrives, the pipeline upgrades without retraining the others.

The DINO confidence distribution in Figure 6 (right) bears this out: cosine scores above 0.7 are rare in absolute count but exist (the high-confidence tail), while the bulk of accepted matches sit between 0.4 and 0.6. This is the expected regime for one-shot prototype matching on out-of-domain crops, and it justifies the modest `DINO_THRESHOLD = 0.4` we use — raising the threshold would lose recall faster than it would gain precision.

5.2 Negative result: hand identity from box geometry (RQ3)

We hypothesized that infant (“self”) hands would be systematically larger in pixel area than caregiver (“other”) hands within the egocentric frame, because the infant wears the camera and their hands are usually closer. We attempted **five variants** of geometric thresholding: (V1) global area threshold, (V2) per-frame relative size, (V3) area + aspect ratio, (V4) area + AR + vertical position, and (V5) adding mean-skin-tone intensity. *All five failed.*

Table 4: Failure modes of geometric infant/caregiver discrimination.

Failure mode	Cause
Adult close to camera = “infant” Infant far from camera = “adult”	Parent leans over toy table; pixel scale matches infant. Infant fully extends arm; pixel area drops below threshold.
Per-frame larger $\neq$ infant	Adult reaching for toy infant is also touching the larger box.
AR collapses across populations	Grip posture dominates AR variance, not anatomy.
Vertical position unreliable	Caregivers reach from many directions.
Skin-tone signal washed out	Auto-exposure dominates intra-class skin variance.

Failure modes (Table 4).

Root cause. The fundamental issue is a **confounding-variable problem**: pixel size measures *distance from camera*, not *anatomical identity*. Both populations occupy a wide, overlapping range of distances (~ 15 cm to ~ 1 m): an infant’s fully extended arm puts their own hand at small pixel scale; a caregiver leaning over the toy table puts theirs at large pixel scale. Aspect ratio collapsed across populations because grip posture (the dominant manipulation in this dataset) overrides anatomy. Vertical position failed because caregivers reach in from many directions. Skin-tone intensity was washed out by auto-exposure on the egocentric rig.

Crisply: *identity does not survive the projection*. Recovering it requires an *appearance-based* model on the cropped pixels, not a geometric heuristic. Three correct fixes are in scope for future work: a small supervised CNN classifier on labeled hand crops, a SimCLR-pretrained encoder + linear probe, or a forearm-strip context classifier.

5.3 Negative result: SAM2 as a standalone object proposer

We hypothesized that SAM2’s class-agnostic mask generator + DINOv2 prototypes would replace supervised object detection entirely. We swept six SAM2 / DINO parameters (`points_per_side`, `pred_iou_thresh`, `stability_score_thresh`, `MAX_BOXES`, `DINO_THRESHOLD`, crop padding); no combination produced a usable signal. We documented eight failure modes (over-segmentation,

small-object miss, hand-object merging, hand-object splitting at occlusion, ranking metric anti-correlated with goal, DINO false matches on smooth backgrounds, frame-to-frame label flicker, compute cost).

Root cause. The diagnosis is **structural, not parameter-tunable**: SAM2’s `pred_iou` and `stability_score` reward smooth, large, well-defined regions, which is the opposite of what held toys look like in egocentric video (small, irregular, partially occluded). Without a “where is the interaction” prior, the top- K masks are dominated by rugs, walls, and table surfaces. Threshold sweeps just trade one symptom (mask flood) for another (only background masks). Figure 11 shows a representative failure: of 10 SAM2 region masks generated on a single frame, all 10 receive a low-confidence `Bed` label from DINO ($\cos \in [0.29, 0.31]$); the held object never enters the top- K .

SAM2 failure mode: top- K masks are dominated by background regions classified as ‘Bed’ or ‘unknown’ at low cosine similarity. The held object never makes the top- K .



Figure 11: SAM2 standalone failure on Baby 25190 frame 127. *Top*: raw frame and SAM2+DINO annotated output. *Bottom*: four of the 10 SAM2 region crops produced for this frame. All 10 regions receive the same DINO label (`Bed`, cosine ≈ 0.30) — a known false-positive sink class that we exclude in the production YOLO+DINO path. Tightening or relaxing SAM2 thresholds did not recover the held object; the mask-quality ranking metric is structurally biased toward smooth background regions.

Ablation: YOLO+DINO vs. SAM2+DINO on the same footage. To quantify the structural difference, we compare both pipelines on the same Day-4 video (Baby 25190; Table 5). The YOLO+DINO production pipeline emits 3,320 hand-anchored detections, 230 of which receive a class label above the DINO threshold; the labels concentrate on plausible toy classes (Spoon, Cups, Lion, Avocado, Spatula). The SAM2+DINO ablation pipeline, in contrast, emits 171,212 region detections on the same video — $\sim 51\times$ more — but 78.8% of them are labeled `Bed` (the known false-positive sink), leaving only $\sim 21\%$ usable, at substantially lower mean confidence (0.31 vs. 0.49). The remaining “usable” SAM2 labels are dominated by spurious matches to rare prototypes (`Oven mitt`, `Raccoon`) that do not appear in YOLO+DINO’s top-5 at all — consistent with smooth background regions producing semantically meaningless DINO embeddings.

Table 5: YOLO+DINO vs. SAM2+DINO on the same Baby 25190 Day-4 footage.

Metric	YOLO+DINO (production)	SAM2+DINO (ablation)
Total detections	3,320	171,212
Labeled (above DINO threshold)	230 (6.9%)	171,212 (100%)
Of which Bed -class (FP sink)	0 (excluded by design)	134,842 (78.8%)
Usable (non- Bed , non- unknown)	230 (6.9% of total)	36,370 (21.2% of total)
Mean confidence	0.49 (DINO cosine)	0.31 (DINO cosine)
Top-5 labels	Spoon, Cups, Lion, Avocado, Spatula	Bed, Lion, Oven mitt, Ball, Raccoon

This negative result directly motivated the final architecture: we replaced SAM2 with the YOLO hand detector as the spatial prior, and the dynamically padded crop replaced mask generation entirely. The cost is zero—we already had the YOLO detector. The benefit is decisive: the held object now reliably falls inside the crop being embedded, the **Bed** false-positive sink disappears, and the labels that survive the DINO threshold are semantically plausible toy classes (Table 5, left column).

5.4 Limitations

1. **DINO localizes to the hand region, not the object boundary.** The pipeline correctly identifies that a target object is present in the crop, but the resulting “object box” is the padded hand box, not a tight box on the object. Results should be read as **hand-region object associations**, not pixel-precise object localization.
2. **Hand identity (infant vs. caregiver) is unsolved.** Per-infant statistics are valid as “objects co-occurring with hands in this infant’s egocentric workspace,” not as “objects this infant grasped with their own hands.”
3. **Global DINO threshold.** A single `DINO_THRESHOLD = 0.4` is applied across all 23 classes. Per-class calibration (e.g., from a small validation set) would likely raise recall on visually similar classes and reduce false positives on rare classes. The Day-4 confidence distribution in Figure 6 (right) shows that few detections sit near the threshold for most classes, but the **Cups** prototype dominates many infants’ counts and likely absorbs visually similar near-classes (bowls, lids).
4. **Two-handed double-counting.** Each detected hand contributes its own object label; a child holding one ball with two hands counts as two ball-frames. Appropriate for *contact-time* analyses, inappropriate for *object-presence* counts.
5. **Five subjects; uneven sample sizes.** Per-infant detection counts vary by $\sim 28\times$ (Baby 25190 with $n = 230$ vs. Baby 25602 with $n = 6,562$). The χ^2 result establishes that the *pipeline* recovers between-subject variation, not that the variation generalizes to a population. The small- n infant in particular should be interpreted with caution.
6. **Metrics are on a single training run.** The reported numbers are from one 100-epoch run. We did not perform multi-seed averaging; performance variance across seeds is unmeasured.

5.5 Broader implications

The cross-cutting lesson from §5.2 and §5.3 is that **off-the-shelf general-purpose tools fail in egocentric infant video for the same structural reason**: the domain has a specific learned

prior (where hands are, what graspable objects look like) that the general tool lacks. Adding the missing prior—a YOLO hand detector, an appearance classifier on the crop—turns each problem into a tractable one. We were able to do this for the SAM2 case because we already had the hand detector; we were not able to do it for the hand-identity case because we did not budget for the additional labeled data. This pattern generalizes to other egocentric-CV problems: budget for the domain-specific prior up front; do not try to recover it from general-purpose features alone.

6 Conclusion

We built and validated an end-to-end pipeline for egocentric infant hand detection and one-shot in-hand object–interaction labeling. The hand detector reaches $\text{mAP}@50 = 0.96$ ($\text{mAP}@50-95 = 0.72$) on held-out footage from 5 infants and generalizes to new environments. Pairing the detector with DINOv2 prototype matching on a dynamically padded crop enables object–interaction labeling across 23 toy classes without per-class supervised training. Per-infant interaction distributions are highly statistically distinct (Pearson $\chi^2 = 14,877.81$, $\text{dof} = 88$, $p < 10^{-3}$), recovering meaningful behavioral variation that is interpretable for downstream developmental analyses.

Two negative results—geometric infant/adult hand discrimination and SAM2 standalone object proposing—were diagnosed in detail and directly motivated the final architecture. We argue that the structural pattern they share (general-purpose tool, missing domain-specific prior) is the central design lesson of the project.

Future work.

1. **Object-boundary localization** within the hand crop, e.g., a second SAM2 pass *prompted by the YOLO hand box* to recover tight object masks.
2. **Hand identity** via a small appearance CNN or SimCLR + linear probe trained on labeled infant-vs-caregiver crops; the SimCLR scaffolding is already in the repository.
3. **Per-class threshold calibration** for DINO cosine matching, prioritizing visually similar pairs (cups/bowls/lids).
4. **Scale-up**: applying the pipeline to a larger cohort (≥ 30 infants) to test population-level developmental hypotheses about object preference and word learning, building on Bambach et al. [1].

Impact. The pipeline lowers the cost of egocentric infant analysis from manual frame-by-frame coding (hours per minute of footage) to automated processing (real-time on a GPU). For developmental psychology specifically, it makes feasible studies of object exposure and interaction at scales previously gated by annotation labor.

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A Reproducibility

Single-command reproduction from a clean checkout:

```
# Stage 1-4: extract frames, MediaPipe, YOLO labels, dedup
bash bash/run_pipeline_part1.sh

# Manual CVAT review on dataset/labels/train/

# Stage 5-6: split + YOLOv8m fine-tune
bash bash/run_pipeline_part2.sh # local
sbatch slurm/run_pipeline_part2_slurm.sh # HPC

# DINO prototypes (one-time)
python scripts/train_dino.py

# End-to-end YOLO + DINO inference
sbatch slurm/run_yolo_dino.sh

# Regenerate all paper figures from data
python scripts/make_figures.py
```

All hyperparameters: `scripts/config.py`. Training-run weights: `runs/detect/train5/weights/best.pt`.
Prototype bank: `prototypes.pth` (24 classes constructed; “Bed” excluded at load → 23 active classes).